# Giray Coskun

**Software | DevOps Engineer**
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## Summary

Software Engineer with experience across backend development, cloud infrastructure, and DevOps automation. Built Java/Python services, AWS-based deployments, and CI/CD pipelines for production systems in ERP and e-commerce domains. Hands-on with distributed systems, self-hosting, and release automation, with additional full-stack delivery using React and TypeScript.

## Core Skills

**Programming:** Python, Java, Go, TypeScript, JavaScript, SQL
**Backend:** REST APIs, Microservices, Distributed Systems, Event-Driven Workflows, Spring Boot, FastAPI, Flask
**Cloud & DevOps:** AWS, Docker, Kubernetes, Terraform, CI/CD, Linux, Nginx, GitHub Actions
**Databases:** PostgreSQL, MySQL, MongoDB
**Frontend:** React, Vue.js, Vite, Tailwind CSS
**Spoken Languages:** English (Full professional proficiency), Turkish (native), German (B1)

## Experience

**DevOps Engineer** Sep 2023 – Mar 2024
*RedRose Technology* Istanbul, Turkey

- Managed AWS infrastructure and production deployments for Java/Python ERP services used by 200+ NGOs.
- Implemented CI/CD pipelines using Docker to standardize build, release, deployment, and rollback workflows.
- Automated deployment processes and routine operations on Linux-based cloud environments with production support ownership.
- Defined release guardrails with development teams, including deployment checklists and rollback-ready procedures.

**Cloud Consultant** Jun 2022 – Sep 2022
*Skyloop* Istanbul, Turkey

- Designed and implemented AWS cloud infrastructure solutions for client migration projects with Terraform.
- Built cloud-native data workflows with AWS Lambda, AWS Kinesis, and Amazon S3 using Python.
- Prepared migration architecture notes and delivery documentation to support client handover and operations readiness.

**Backend Developer Intern** Jun 2021 – Sep 2021
*Migros E-commerce* Istanbul, Turkey

- Developed backend services for e-commerce systems using Java and Spring.
- Supported service containerization and data operations with Docker and MySQL.
- Contributed to API integration validation and staged backend deployments across development environments.

**Software Engineer Intern** Jun 2020 – Sep 2020
*Havelsan* Istanbul, Turkey

- Conducted UAV mission planning research with linear programming genetic algorithms in Go.

**MLOps Engineer Intern** Feb 2020 – Jun 2020
*Spiky AI* Buffalo, New York (Remote)

- Implemented data pipelines and model training workflows for video sentiment analysis using Python and PyTorch in AWS Infrastructure.

## Projects

**Automated Reasoning API** GitHub
Developed a backend service with microservice and queue-based architecture for reasoning query processing.

**Home Server** GitHub
Built a home server with a dashboard via React, Vite, TypeScript, and Nginx to manage self-hosted services, status pages, and operational visibility.

- Operated 18+ self-hosted services across Docker and systemd, including Grafana, Home Assistant, and Jellyfin.
- Configured Nginx reverse-proxy and route mapping for local vs external delivery paths, enabling controlled environment-specific releases.

**Webhook Server** GitHub
Built a Python/Flask webhook server to trigger deployment scripts from GitHub webhooks and password-protected manual endpoints.

- Implemented HMAC verification, background jobs, project locks, and Swagger docs.
- Automated self-hosted project updates through shell scripts and systemd workflows for services including the CV site and personal dashboard.

**Encryption in Assembly** GitHub
Implemented a custom AES-like encryption algorithm in x86-64 assembly language.

## Education

**Graduate Coursework in Informatics** Apr 2024 – Mar 2026
Technical University of Munich, Munich, Germany
*Coursework:* Cloud Data Processing, Distributed Systems, Machine Learning
*Focus:* Data engineering and scalable systems; program not completed

**B.Sc. Computer Science and Engineering and Economics Double Major** Sep 2017 – Jun 2023
Sabanci University, Istanbul, Turkey
Minor in Mathematics, GPA: 3.78/4 in Computer Science, 3.66/4 in Economics
Ranked among the top 500 in Turkey's university entrance examination (2017).

## Certification

**AWS Certified Solutions Architect – Associate** Issued: Aug 2022
**TOEFL iBT: 110/120** Issued: Oct 2022

## Publications

**Using Unified Combinatorial Interaction Testing for MC/DC Coverage**
Published in IEEE International Conference on Software Testing, Verification and Validation (ICST), 2022.
IEEE Xplore Link

**A Combinatorial Interaction Testing-Based Daily Build Process**
Published in Journal of Systems and Software, Volume 190, 2022.
ScienceDirect Link

**Unsupervised Adaptation of DNN for Brain-Computer Interface Spellers**
Published in Signal Processing and Communications Applications Conference (SIU), 2022.
DOI Link